Software Engineering

The Software Process(Spiral&CMMI)

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Spiral Model

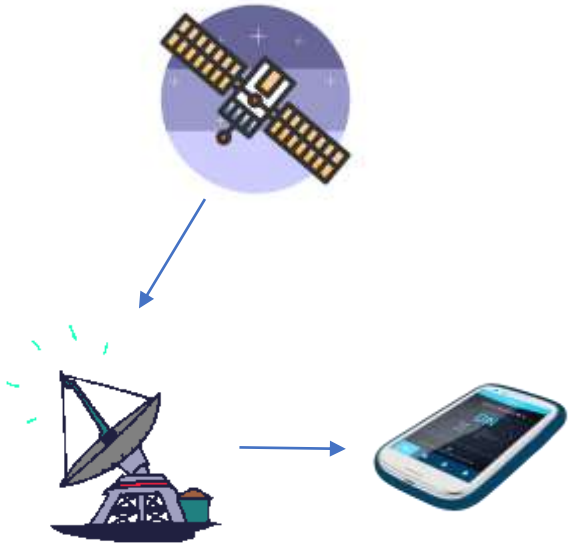
- Rarely Used but an important model
- Its a model that works for projects with unlimited budget, time and projects that has huge risk factors.
- Example, making a heavy lift system for space shuttle and international space station.



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Galaxy Inc.

- They wanted to send 6 dozens satellite in space and build a satellite based cellular system.
- So that remote places like even in Antarctica where you don't have any BTS(mobile tower) you can still be able to communicate using your cell.
- You are never out of network.



As you can see for this project:

- Risk were enormous
- Needs a huge budget
- No published materials or experienced worker
- Risks will be coming and identified once the project kicks off

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Spiral Model Formal Definition

- The **spiral model** is a **risk-driven** process model generator for software projects. Based on the unique risk patterns of a given project, the spiral model guides a team to adopt elements of one or more process models, such as incremental, waterfall, v-model or iterative prototyping.
- This model was first described by **Barry Boehm** in his 1986 paper "A Spiral Model of Software Development and Enhancement".



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Spiral model Figure



- Here is the image depicting spiral model.
- As you can see spiral loops showing phase by phase development.
- You can see we are doing risk analysis in every phase, planning and keep building prototype until we reach our goal.

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Spiral model sectors

- Objective setting
  - Specific objectives for the phase are identified
- Risk assessment and reduction
  - Risks are assessed and activities put in place to reduce key risks
- Development and validation
  - A development model for the system is chosen which can be any of the generic models
- Planning
  - The project is reviewed and next phase of the spiral is planned



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Spiral model usage

- Spiral model has been very influential in helping people to think about iteration in software processes and introducing the risk-driven approach to development. In practice, however as mentioned, the model is rarely used as published for practical software development.
- So if you are a project manager or lead developer when would you suggest spiral model? It is if you have:
  - Long term project commitment and budget
  - Users and developers unsure of the needs
  - Requirements are complex
  - New product line (a new category product)
  - Significant changes are expected(research and explanation)



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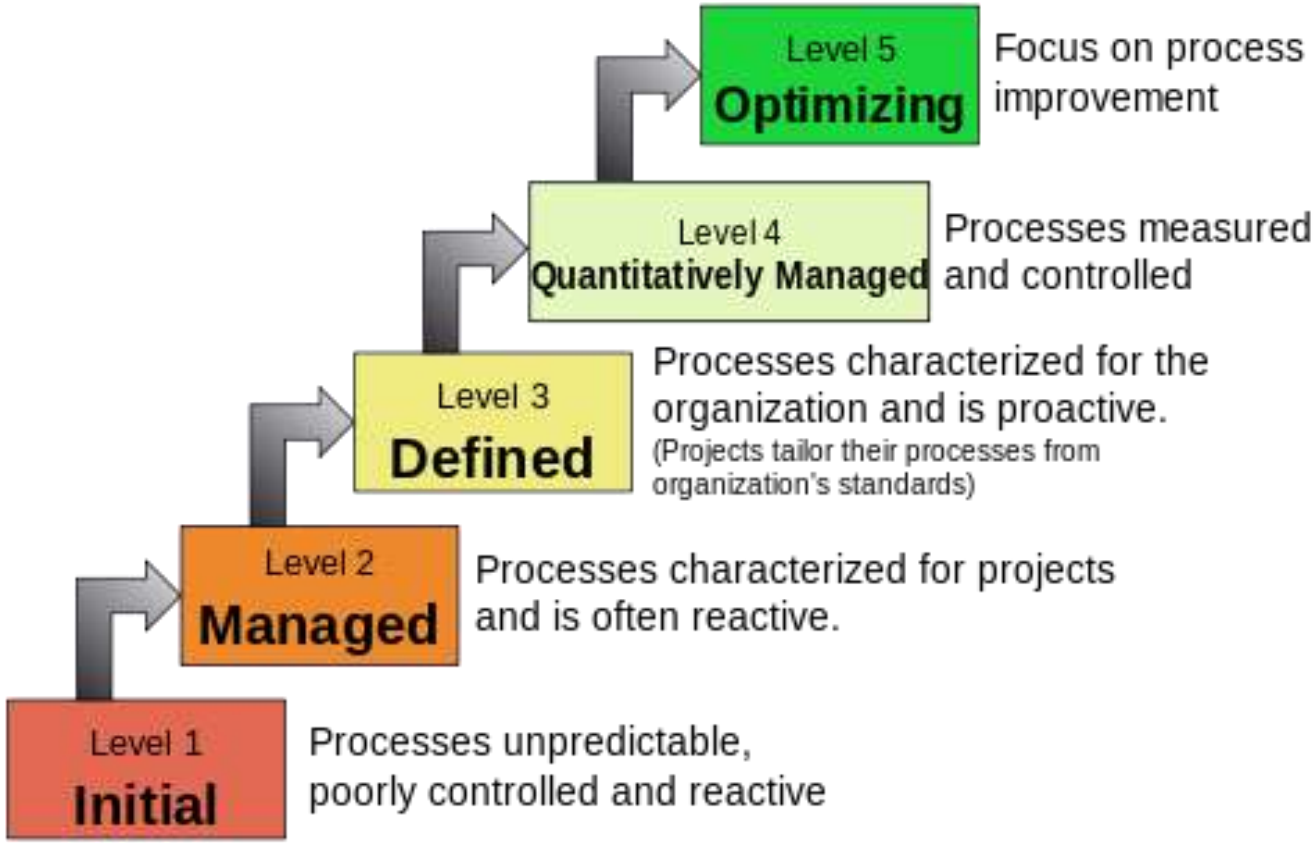
CMMI

- The Capability Maturity Model Integration (CMMI) is a process and behavioral model that helps organizations streamline process improvement and encourage productive, efficient behaviors that decrease risks in software, product and service development. It measures the maturity of the organization in terms of Software Development Process.
- It is developed by CMU
- This process is mostly a requirement if you want to get a contract for software development in US govt organization.
- In this model work is divided in such a way so that you have different maturity level of a system you are building.
- It divided in 5 maturity level and you need to improve the system until you reach level 5



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Characteristics of the Maturity levels



- Once you reach level 5 that does not mean the end of your system.
- It means now the system is full proof, it just need regular maintenance nothing else.

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CMMI Model – Maturity Levels :

Maturity level 1 : Initial

1. Processes are poorly managed or controlled.
2. Unpredictable outcomes of processes involved.
3. Ad hoc and chaotic approach used.
4. Lowest quality and highest risk.

Maturity level 2 : Managed

1. Requirements are managed.
2. Processes are planned and controlled.
3. Projects are managed and implemented according to their documented plans.
4. This risk involved is lower than Initial level, but still exists.
5. Quality is better than Initial level.

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Maturity level 3 : Defined

1. Processes are well characterized and described using standards, proper procedures, and methods, tools, etc.
2. Medium quality and medium risk involved.
3. Focus is process standardization.

Maturity level 4 : Quantitatively managed

1. Quantitative objectives for process performance and quality are set.
2. Quantitative objectives are based on customer requirements, organization needs, etc.
3. Process performance measures are analyzed quantitatively.
4. Higher quality of processes is achieved.
5. Lower risk

Maturity level 5 : Optimizing

1. Continuous improvement in processes and their performance.
2. Improvement has to be both incremental and innovative.
3. Highest quality of processes.
4. Lowest risk in processes and their performance.

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Level	Focus	Process Area	
5 Optimizing	Continuous Process Improvement	•Organizational Performance Management	•Causal Analysis & Resolution
4 Quantitatively Managed	Quantitative Management	•Organizational Process Performance	•Quantitative Project Management
3 Defined	Process Standardization	•Requirements Development •Technical Solutions •Product Integration •Verification •Validation •Organizational Process Focus	•Organizational Process Definition •Organizational Training •Integrated Project Management Risk Management •Decision Analysis & Resolution
2 Managed	Basic Project Management	•Requirements Management •Project Planning •Project Monitoring & Control •Supplier Agreement Management	•Measurement & Analysis •Process & Product Quality Assurance •Configuration Management
1 Initial			

- Pause and go through this chart that will give you detailed Idea of what happens in each maturity levels

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